

StreamingEngine MII Project Survey

SAP MII Workbench project analysis — internal technical reference | 2026-07-23

1. Overview

Project: **StreamingEngine** | SAP MII version: **15.5.1.0** | Export roots: **Catalog, WEB, Meta-Inf** | Total files: **861**

2. Structure Confirmed

- **Catalog**: 25 modules, each with Query/ (.tqsq SQL, .tqxc Xacute, .tqpc PCo/tag) and Transaction/ (.trx) subfolders — 705 catalog items total.
- **Web**: SAPUI5 app (Component.js, manifest.json, 48 views/fragments, 35 controllers, models, i18n) — not legacy Visual Composer.
- **Meta-Inf**: ProjectInfo.xml confirms MII 15.5.1.0. Roles.xml and jobsconfig.xml are empty — role mappings and job schedules live in MII Admin, not in this export.

3. Web-to-Catalog Binding Mechanism (Verified)

No OData. Every controller calls
/XMII/Illuminator?QueryTemplate=StreamingEngine/<Module>/Query|Transaction/<Name> via raw \$.ajax, reading result.Rowsets.Rowset[0].Row[...]. 141 such references were checked against the Catalog tree — **139 resolve, 2 do not** (see Red Flags, item 1).

4. Architecture Facts for Troubleshooting

- Error handling is **Catch-block based** (try/catch), not classic Success/Failure link arrows (MII 15.x pattern).
- All SQL queries (440 files) hit one data source: **Z_STREAMING_ENGINE**.
- Xacute queries (78 files) are wrapper actions that directly execute a named Transaction.
- Alerts fire from inside Transactions (AlertNotification/Transaction/AlertTagsStatusCheck_Job.trx), never from the web page directly.

5. Module-by-Module Map

Module	Purpose	Query/ Trx	Key Items	WEB View(s)
Common	Shared lookups (plants, sources, connections, PCo destinations, flow types)	13 / 3	PlantListSelectQuery, SourceListSelectQuery, SQLCommandQuery	Used by almost every view (FlowMonitoring, JobsMonitoring, QueueMonitoring, etc.)
DataFlow	Flow monitoring, flow definitions (Contextual/Archive/File/External), extended transactions	84 / 44 (largest)	FlowMonitoringListSelectQuery, FlowMonitoringListAggregateQuery, ExternalFlowListSelectQuery	FlowMonitoring.view.xml, ContextualFlows/Information, ArchiveFlows/Information, FileFlows/Information, ExternalFlows/Information

Module	Purpose	Query/ Trx	Key Items	WEB View(s)
JobEngine	Core scheduling/execution engine: job queue mgmt, catch-up/retry, per-flow-type execution	71 / 22	EngineExecution_Job.trx, EngineExecution_EnergyJob.trx, JobMonitoringListSelectQuery, JobEndDateUpdateXacuteQuery, JobCatchupReloadXacuteQuery	JobsMonitoring.view.xml, QueueMonitoring.view.xml
PCoAgent	PCo agent inventory/monitoring/restart, tag-mapping sync	19 / 7	AgentMonitoringListSelectQuery, AgentRestartXacuteQuery	PcoMonitoring.view.xml
PCoDestination	PCo destination lookup only	1 / 0	Single select query	Referenced by DataDestination views
TagCatalog	Tag inventory, OPC UA discovery, tag/digital-app assignment	76 / 26	OPC UA group/tag list PCo queries, digital-app tag update	TagCatalog.view.xml, TagInformation.view.xml, TagUpload(1).view.xml
TagWatch	Watch/history capture for tags, visualization limits	11 / 3	TagHistoryTransaction.trx, TagWatchMonitoring_Job.trx, TagHistoryPcoQuery.tqpc (OSISoft PI)	Feeds TagInformation view
BatchInformation	Batch catalog import/download, equipment lists from OSISoft PI AF / Marcy SQL	42 / 13	BatchCatalogImportXacuteQuery, per-source EquipmentList_* PCo queries	BatchCatalog.view.xml, BatchCatalogUpload.view.xml, BatchInformation.view.xml
DataSource	Plant data-source config (Kepware/OPC UA/PI), CRUD + connection testing	34 / 11	readFolder.tqpc (Kepware OPC UA), TagsByFilterRetrieveQuery.tqpc (OSISoft PI)	DataSources.view.xml, DataSourceInformation.view.xml
DataDestination	Where processed data is sent (API/webservice/SFTP/PCo)	25 / 8	DestinationHTTPParam*, TestConnectionDestinationXacuteQuery	DataDestinations.view.xml, DataDestinationInformation.view.xml (has a dead query reference — see Red Flags)
ManualImport	Manual file/data import flows	34 / 7	—	ManualImport.view.xml, ManualImportUpload.view.xml
HistoricalUpload	Historical/backfill upload of tag data	11 / 2	UserProfileEmailListSelectForHistoricalUploadQuery (in AlertNotification, cross-module)	Fragments: HistoricalDataUpload / HistoricalUploadReport
InitialLoad	One-time initial load of flows/tags on flow creation	15 / 1	Referenced from EngineExecution_Job.trx (InitialLoadFlowDeleteQuery, etc.)	Fragments: InitialLoadDateRange / InitialLoadReport
AlertNotification	Bad-tag/job/queue/PCo-agent alerting via email	11 / 3	SendEmailAlertTransaction.trx, AlertTagsStatusCheck_Job.trx	No direct view — backend-only, polled via NotificationSystem
NotificationSystem	In-app notification delivery/polling	6 / 1	DeleteDeliveredNotifsAndCheckQueueXacuteQuery	Polled app-wide from BaseController.js
AuditLog	Change/audit history, restore	17 / 5	AuditLogHistorySelectQuery, FullAuditLogDownloadTransaction.trx	AuditLog.view.xml
DigitalApp	Digital app assignment to tags	14 / 2	—	DigitalApps.view.xml
PlantHierarchy	Plant/hierarchy structure	12 / 3	—	Supports PlantInformation
PlantParameter	Per-plant configuration parameters (job durations, thresholds)	10 / 2	—	PlantParameters.view.xml

Module	Purpose	Query/ Trx	Key Items	WEB View(s)
ReferenceUpdate	Reference-data update utility	2 / 2	—	ReferenceUpdate.view.xml
TagName	Tag naming/renaming utilities	6 / 1	—	Supports TagCatalog/TagInformation
UnitOfMeasure	UoM lookups	4 / 1	—	Supports TagInformation
UserManagement	User admin	9 / 2	—	UsersManagement.view.xml
UserProfile	Roles, email addresses for alerts	15 / 4	UserProfileEmailListSelectQuery (used by AlertNotification)	UserInformation.view.xml, ProfilePage.view.xml
LaunchPad	App landing page tiles	3 / 1	—	Launchpad.view.xml

6. Red Flags Found

1. Confirmed dead reference / live bug

`DataDestinationInformation.controller.js:480` calls a query (`HistoricalUploadDestByDestinationSelectQuery`) that does not exist in Catalog. There is a *"/ TODO: replace with your real query"* comment directly above it, and both the success and error handlers are no-ops, so it fails silently on every call. Likely cause of a broken Historical Upload dropdown on that page.

2. Missing var that = this

In `JobsMonitoring.controller.js:290` and `FlowMonitoring.controller.js:144` — if either query returns a `FatalError`, a JS `ReferenceError` occurs instead of the intended error message. In `JobsMonitoring`, the busy dialog never closes on that path.

3. Hardcoded environment hostnames

Baked into ~10 controllers (`sapdqnjc.pharma.aventis.com`, `sapwdawsemea.pharma.aventis.com`) — will not auto-follow a DEV → QA → PRD promotion.

4. Mixed dev/test/prod PCo connection names

Dev/test connection names (e.g. `SE_PCo_TEST_KEPWARE`) sit alongside production ones in the same `TagCatalog` folder.

5. Client-side-only role gating

`STREAMING_ENGINE_ADMIN/USER` role checks are enforced client-side only in the JS reviewed; actual enforcement should be the `ReaderRoles/writerRoles` attributes on Catalog items.